

$$= \frac{1}{2R} - \left(\frac{1}{R} \right) = -\frac{1}{2R}$$

10 C Top: $F_{\text{top}} + mg = mr\omega^2$

$$F_{\text{top}} = 77 [4(2\pi/3.7)^2 - 9.81] = 133 \text{ N (downwards)}$$

Bottom: $F_{\text{bottom}} = mg + mr\omega^2$

$$F_{\text{bottom}} = 77 [4(2\pi/3.7)^2 + 9.81] = 1640 \text{ N (upwards)}$$

11 D Resultant force towards the axis equals to mg